

An Adaptive Diagonal Criterion for Failure of BCP

Run `fa_banach_001`, lane 19

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Source Problem

The source paper is Jinghao Huang, Karimbergen Kudaybergenov, and Rui Liu, *The ball-covering property of von Neumann algebras and noncommutative symmetric spaces*, arXiv:2311.14257. After proving that a commutative unital C^* -algebra $C(K)$ has BCP/UBCP exactly when K has a countable π -basis, the authors ask for a noncommutative version:

$\{U_\alpha\}_{\alpha \in A}$ of K is called a π -basis (weak basis) for K if every nonempty open subset of K contains some U_α in $\mathcal{U}$ [37, Page 15].

In the setting of commutative C^* -algebra, [25, Theorem 2.1] can be reformulated as follows. Note that [25, Theorem 2.1] only deals with real continuous functions. The complex case is a direct consequence of the real case.

Corollary 3.9. *Let $\mathcal{A}$ be a unital commutative C^* -algebra. Then the followings are equivalent:*

- (1) $\mathcal{A}$ has the BCP;
- (2) the pure state space $\mathfrak{P}(\mathcal{A})$ has a countable π -basis;
- (3) $\mathcal{A}$ has the UBCP.

In particular, there exist commutative C^* -algebras containing no minimal projections but having the BCP. For example, the uniform norm closure of the linear span of Haar functions [24, p.150].

We conclude this section by the following question.

Question 3. *It will be interesting to find a non-commutative version of Corollary 3.9, that is, how to characterize those C^* -algebras $\mathcal{A}$ having BCP/UBCP.*

Idea of the Proof

Several previous packets use the same diagonal escape. A proposed center is nearly normed on a fresh coordinate, block, annulus, or two-sided corner. One then puts a unit vector in that coordinate pointing away from the local image of the center, and glues all chosen coordinates into one global unit vector. The global vector stays outside every ball $B(x_i, \|x_i\|)$.

This packet records the exact form of that mechanism. The primitive ideal space cannot see it: a II_1 factor has one-point primitive ideal space, but its diffuse finite projection structure supplies adaptive two-sided corners.

The Abstract Criterion

For normed spaces E_k , write

$$\ell_\infty(E_k) = \{(u_k) : \sup_k \|u_k\| < \infty\}.$$

Definition 1. Let X be a normed space. We say that X has the adaptive exact diagonal property if for every sequence $(x_i)_{i \geq 1} \subset X$ there are:

- a map $\pi : \mathbb{N} \rightarrow \mathbb{N}$ such that every fiber $\pi^{-1}(\{i\})$ is infinite;
- nonzero normed spaces E_k ;
- contractive linear maps $R_k : X \rightarrow E_k$;
- a linear isometry $\Gamma : \ell_\infty(E_k) \rightarrow X$;
- a null sequence $\eta_k \downarrow 0$;

such that

$$R_k \Gamma((u_j)_{j \geq 1}) = u_k \quad \text{for every } (u_j) \in \ell_\infty(E_j)$$

and

$$\|R_k x_{\pi(k)}\| > \|x_{\pi(k)}\| - \eta_k, \quad k \geq 1.$$

Lemma 1 (one-coordinate escape). Let $E \neq \{0\}$ be a normed space and let $y \in E$. There is $u \in S_E$ such that

$$\|u - y\| \geq \|y\|.$$

Proof. If $y = 0$, take any $u \in S_E$. If $y \neq 0$, choose $f \in S_{E^*}$ with $f(y) = \|y\|$. Choose $v \in S_E$ with $|f(v)| > 0$, and multiply v by a scalar of modulus one, in the complex case, or change its sign, in the real case, so that the resulting $u \in S_E$ satisfies $\operatorname{Re} f(u) \leq 0$. Then

$$\|u - y\| \geq \operatorname{Re} f(y - u) = \|y\| - \operatorname{Re} f(u) \geq \|y\|.$$

□

Theorem 1. If a normed space X has the adaptive exact diagonal property, then X fails BCP.

Proof. We use the equivalent form of BCP saying that S_X is covered by countably many open balls $B(x_i, \|x_i\|)$.

Let (x_i) be an arbitrary countable family in X . Choose adaptive diagonal data for this sequence. For each k , apply Lemma 1 to E_k and

$$y_k = R_k x_{\pi(k)}$$

to choose $u_k \in S_{E_k}$ such that

$$\|u_k - R_k x_{\pi(k)}\| \geq \|R_k x_{\pi(k)}\|.$$

Put

$$z = \Gamma((u_k)_{k \geq 1}).$$

Since Γ is an isometry and $\sup_k \|u_k\| = 1$, we have $z \in S_X$.

Fix i . Along the infinite set $\{k : \pi(k) = i\}$,

$$\|z - x_i\| \geq \|R_k(z - x_i)\| = \|u_k - R_k x_i\| \geq \|R_k x_i\| > \|x_i\| - \eta_k.$$

Letting $k \rightarrow \infty$ through that fiber gives

$$\|z - x_i\| \geq \|x_i\|.$$

Thus z lies outside every open ball $B(x_i, \|x_i\|)$. Since the countable family of centers was arbitrary, X fails BCP. □

C*-Algebra Corner Form

Definition 2. Let A be a C^* -algebra. We say that A has the adaptive exact corner property if for every sequence $(a_i) \subset A$ there are $\pi : \mathbb{N} \rightarrow \mathbb{N}$ with infinite fibers, a null sequence $\eta_k \downarrow 0$, and pairwise orthogonal projections $p_k, q_k \in A$ such that:

- each corner $q_k A p_k$ is nonzero;
- there is a linear isometry

$$\Gamma : \ell_\infty(q_k A p_k) \rightarrow A$$

that satisfies

$$q_k \Gamma((u_j)) p_k = u_k;$$

- the centers are nearly normed on the assigned corners:

$$\|q_k a_{\pi(k)} p_k\| > \|a_{\pi(k)}\| - \eta_k, \quad k \geq 1.$$

Corollary 1. Every C^* -algebra with the adaptive exact corner property fails BCP.

Proof. Apply Theorem 1 with

$$E_k = q_k A p_k, \quad R_k(a) = q_k a p_k,$$

and the gluing map Γ from the definition. □

Von Neumann Corner Gluing

The exact gluing condition is automatic in von Neumann algebras once the left and right support projections are orthogonal.

Lemma 2. Let $\mathcal{M}$ be a von Neumann algebra. Let (p_k) and (q_k) be pairwise orthogonal sequences of projections in $\mathcal{M}$. For every bounded sequence

$$u_k \in q_k \mathcal{M} p_k$$

there is a strong-operator convergent sum

$$u = \sum_{k=1}^{\infty} u_k \in \mathcal{M}$$

with

$$\|u\| = \sup_k \|u_k\|, \quad q_k u p_k = u_k.$$

Proof. Represent $\mathcal{M}$ on a Hilbert space H . For finite $F \subset \mathbb{N}$,

$$\left\| \sum_{k \in F} u_k \right\| \leq \sup_k \|u_k\|,$$

because the initial projections p_k and final projections q_k are orthogonal. More explicitly, for $\xi \in H$,

$$\left\| \sum_{k \in F} u_k \xi \right\|^2 = \sum_{k \in F} \|u_k p_k \xi\|^2 \leq \left(\sup_k \|u_k\| \right)^2 \sum_{k \in F} \|p_k \xi\|^2 \leq \left(\sup_k \|u_k\| \right)^2 \|\xi\|^2.$$

The same estimate on tails shows strong convergence of the net of finite partial sums, and the adjoint sums converge strongly as well. Since $\mathcal{M}$ is strongly closed, the limit belongs to $\mathcal{M}$. The equality $q_k u p_k = u_k$ is immediate from orthogonality, and it gives the reverse norm inequality. $\square$

Corollary 2. *Let $\mathcal{M}$ be a von Neumann algebra. Suppose that for every sequence $(x_i) \subset \mathcal{M}$ there are $\pi : \mathbb{N} \rightarrow \mathbb{N}$ with infinite fibers, a null sequence $\eta_k \downarrow 0$, and pairwise orthogonal projections (p_k) , (q_k) in $\mathcal{M}$ such that $q_k \mathcal{M} p_k \neq \{0\}$ and*

$$\|q_k x_{\pi(k)} p_k\| > \|x_{\pi(k)}\| - \eta_k.$$

Then $\mathcal{M}$ fails BCP.

Proof. Lemma 2 verifies the gluing hypothesis in Corollary 1. $\square$

Exact and Asymptotic Diagonal Examples

The atomless von Neumann proof in Huang–Kudaybergenov–Liu constructs exactly the projections required in Corollary 2: after passing to a σ -finite corner when necessary, their decreasing projection construction produces pairwise orthogonal differences p_k and q_k with

$$\|q_k x_{\pi(k)} p_k\| > \|x_{\pi(k)}\| - \eta_k.$$

Thus the source theorem that nontrivial atomless von Neumann algebras fail BCP is an exact instance of the adaptive corner criterion.

The reduced-product, sigma-unital corona, and Hilbert-module corona packets are asymptotic versions of the same idea rather than literal instances of Definition 1. For a reduced product

$$\ell_\infty(F_n)/c_{\mathcal{I}}(F_n),$$

the proof chooses disjoint positive sets and glues one specifically chosen escape vector on each set. However arbitrary quotient classes on those sets do not provide a canonical ℓ_∞ -sum retraction back into the global quotient: changing representatives on countably many disjoint positive sets need not produce a difference in $c_{\mathcal{I}}(F_n)$. For multiplier coronas the same issue appears analytically: compact or ideal terms vanish only along the chosen tail blocks. The natural next strengthening is therefore an “approximate adaptive diagonal” criterion, with exact retractions replaced by tail-vanishing errors.

Verification and Search Bounds

The run indexes were searched for arXiv:2311.14257 together with BCP, diagonal obstruction, orthogonal corners, von Neumann atomless, reduced products, Hilbert-module block criterion, and sigma-unital coronas. Existing packets contained concrete diagonal arguments but no abstract Banach-space criterion isolating the exact common mechanism.

The proof of Theorem 1 is formal. The von Neumann application uses only standard strong-operator summation over orthogonal corners. The source paper is cited only to identify its atomless von Neumann argument as an instance of the criterion; this packet is not a new proof of the source paper’s full von Neumann classification.

Human Review Focus

Check whether the adaptive exact diagonal property is stated at the right level of generality for future packets. The immediate proof is short; the important review question is how far the exact property can be weakened to cover multiplier coronas and Hilbert-module coronas without redoing each diagonal argument separately.

References

- [1] Jinghao Huang, Karimbergen Kudaybergenov, and Rui Liu, *The ball-covering property of von Neumann algebras and noncommutative symmetric spaces*, arXiv:2311.14257, 2023.